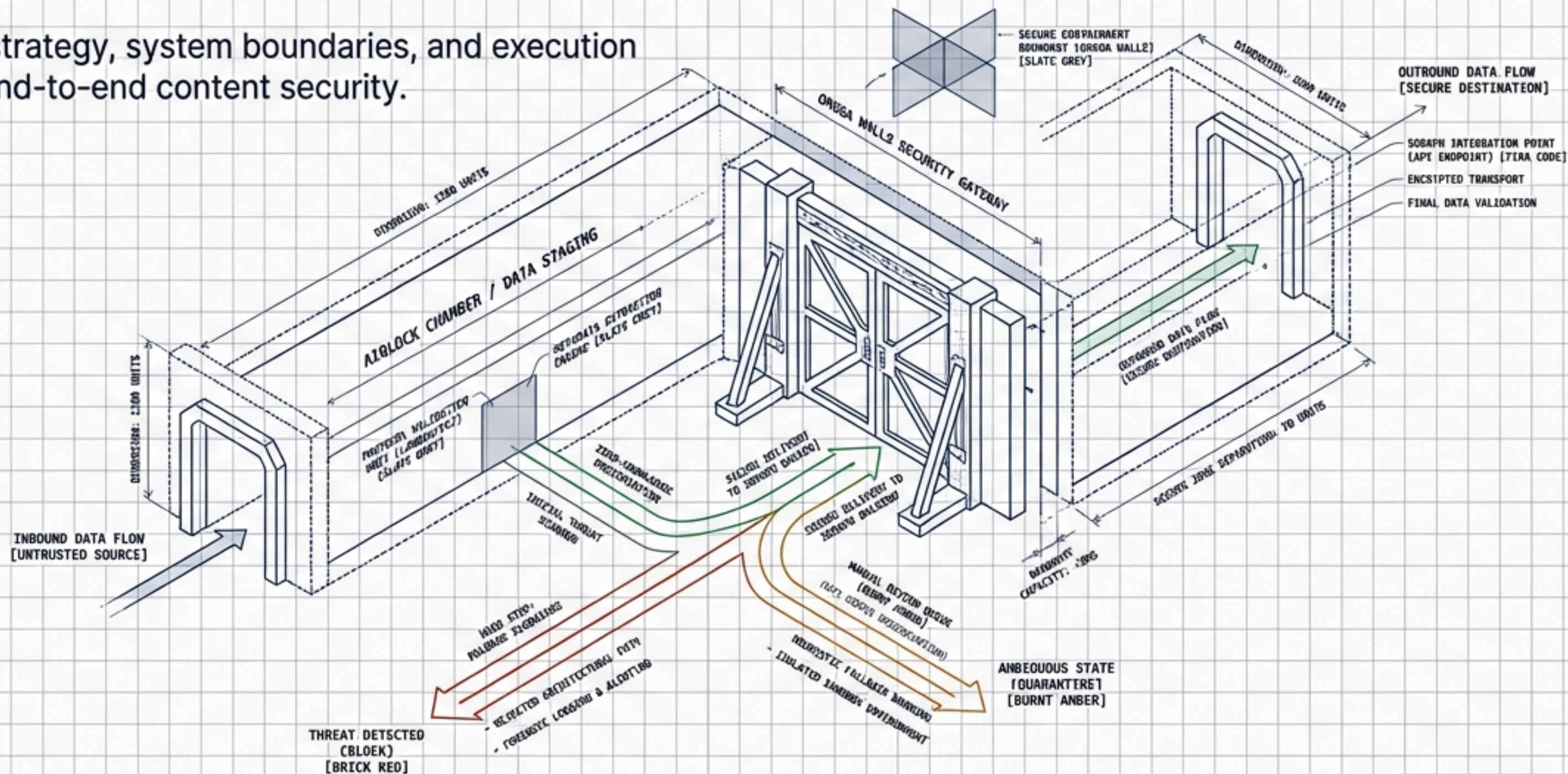


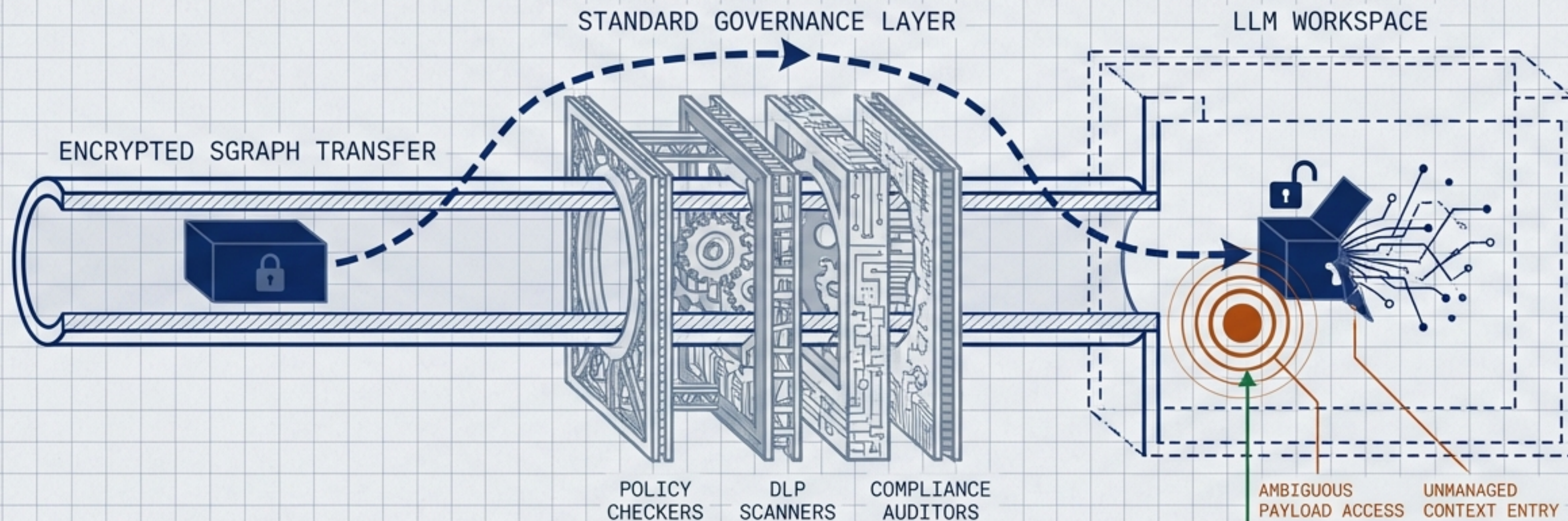
SGraph × Omega Walls MVP Integration Blueprint

Architectural strategy, system boundaries, and execution roadmap for end-to-end content security.



THE ENCRYPTED BLIND SPOT

Encrypted channels guarantee privacy in transit, but create a critical blind spot for content governance.



This integration acts as the security airlock. It intercepts the payload at the exact moment of decryption—before it enters the agentic pipeline.

Threat Vectors at the Point of Ingestion



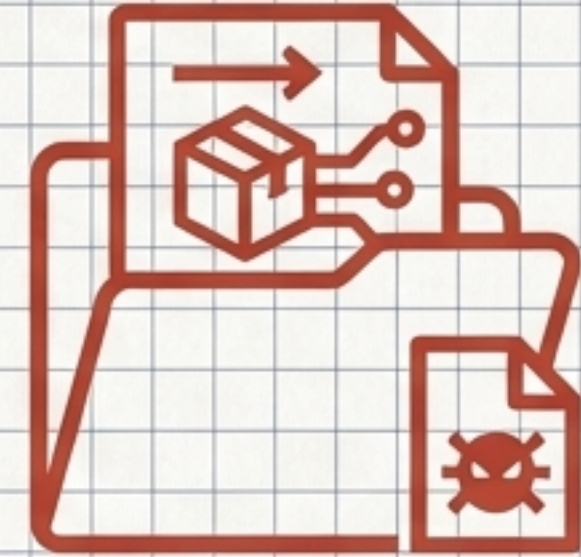
Prompt Injections

Embedded attacks targeting the LLM reasoning engine (e.g., 'Ignore all rules').



Policy Violations

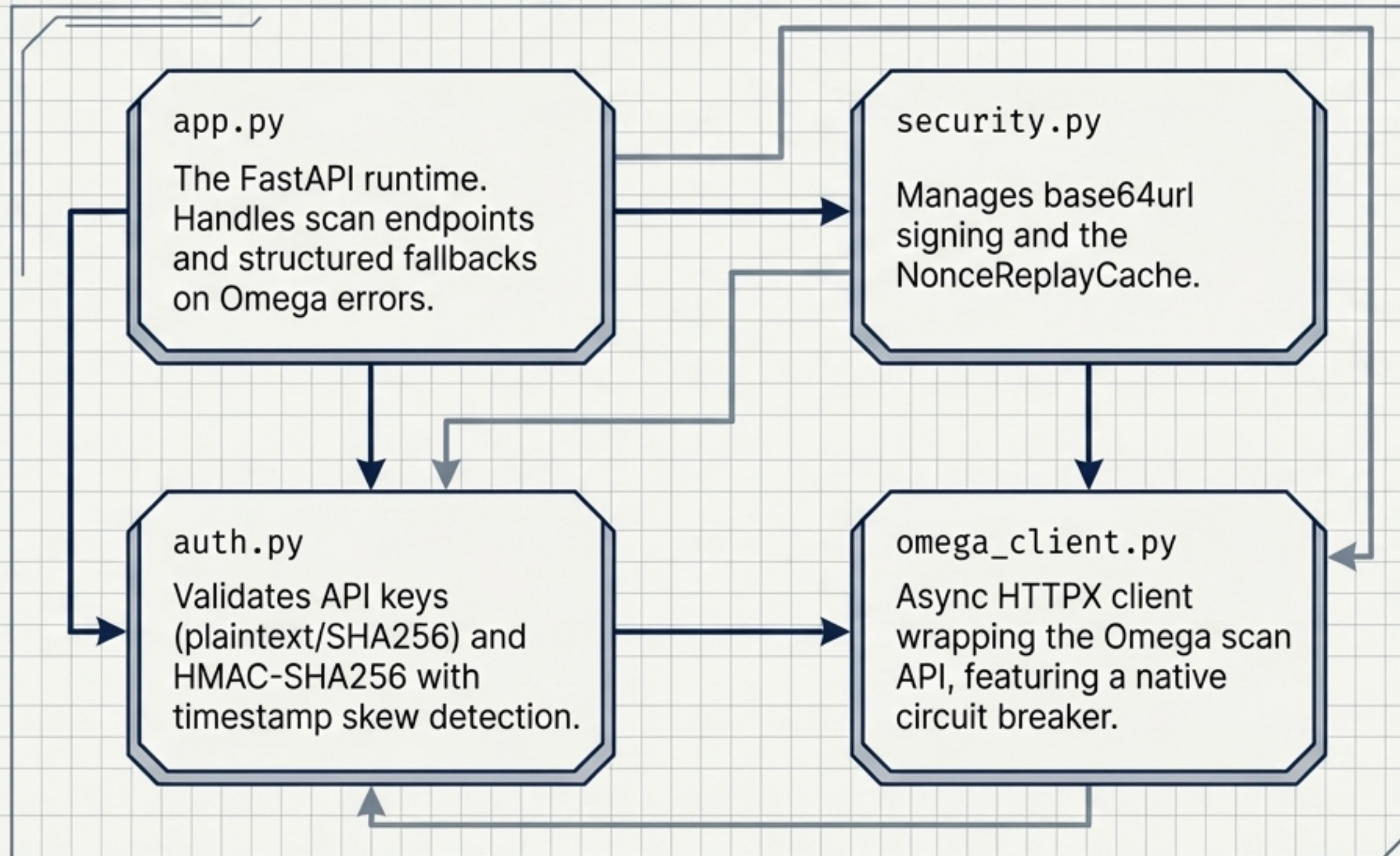
Unauthorised sensitive data entering the LLM pipeline (PII, credentials).



Exfiltration & Malware

Hidden instructions designed to route data to external URLs, or malware disguised as legitimate files.

Current State of the Connector Repository



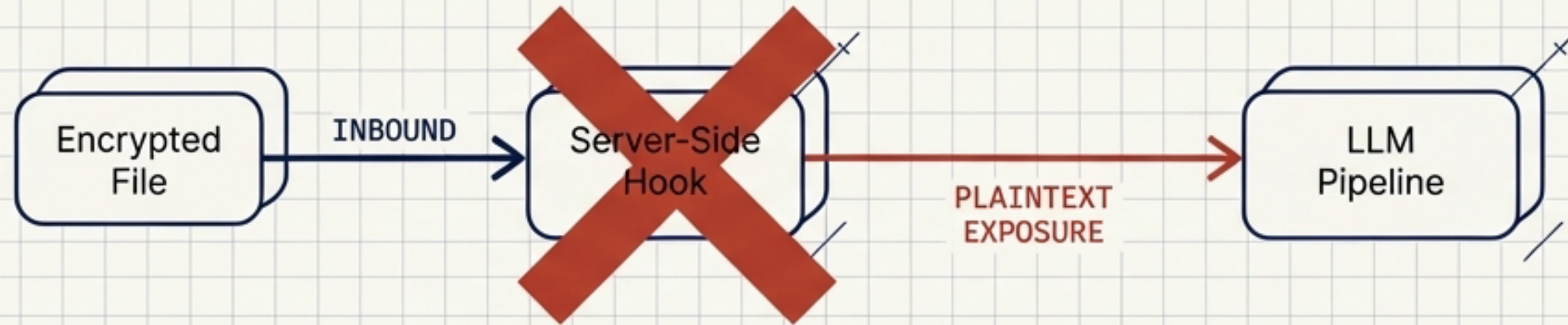
Full e2e compose-gated test suite exists, covering allow/block/quarantine routing.

Compute Architecture and MVP Trade-offs

System Component	AWS Deployment Model	MVP Status
Connector FastAPI	AWS Lambda (Mangum)	✅ Drop-in ready.
Omega Scan Service	EC2 / Fargate	❌ Requires persistent compute (incompatible with Lambda cold starts).
Nonce Replay Cache	Lambda	❌ Broken. In-memory dict fails across concurrent instances. Mitigation: Disable HMAC for MVP.
Circuit Breaker State	Lambda	⚠️ Degraded. State not shared across instances. Mitigation: Acceptable MVP gap, quarantine fail-safe still triggers.

Preserving Zero-Knowledge Encryption

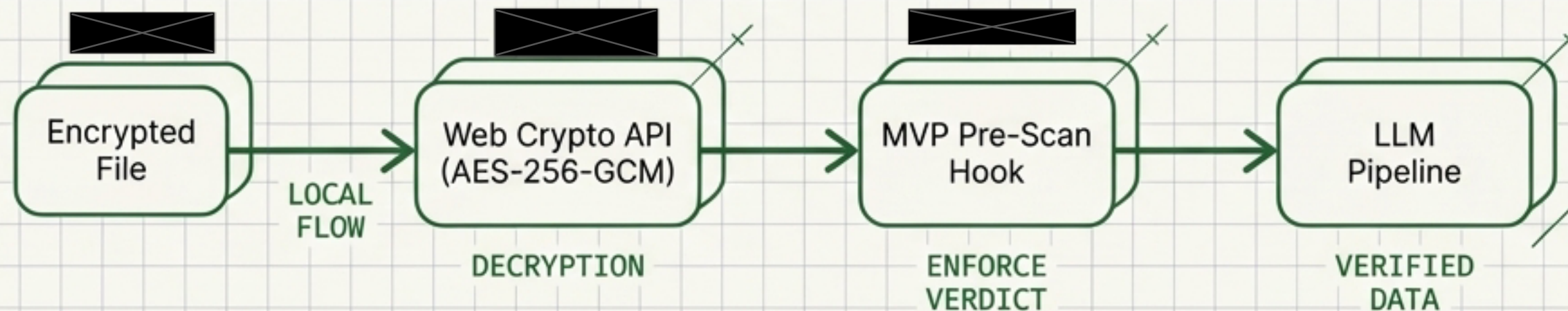
Server Environment



The Conflict:

The initial integration specified a server-side hook. This directly collides with SGraph's core guarantee: the server must never handle plaintext.

SGraph Workspace UI (Browser Boundary)

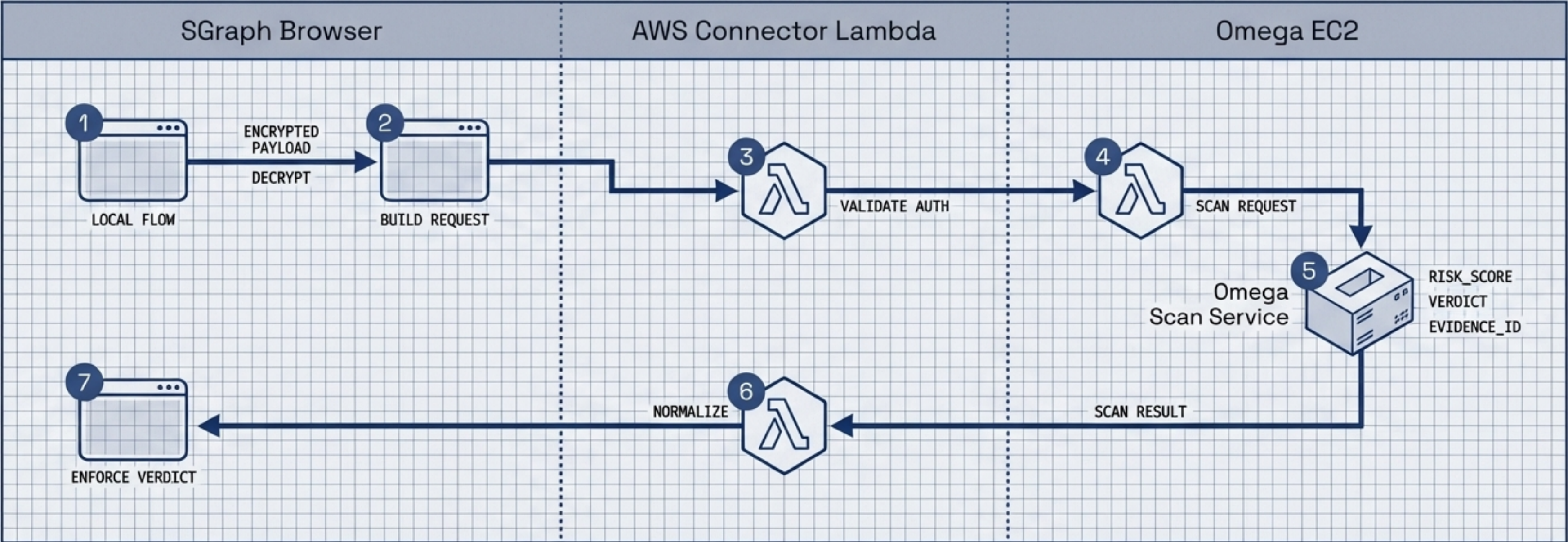


The MVP Resolution:

The pre-scan hook must live in the SGraph Workspace UI.

The browser decrypts the payload, calls the connector, and enforces the verdict before passing data to the LLM.

The End-to-End Pipeline



Legend: Data Pathway

1: SGraph decrypts payload locally (Browser).

2: UI builds connector request with tenant_id, request_id, and file_base64.

3: `POST /v1/scan/attachment` sent to Connector Lambda.

4: Connector validates auth and forwards to Omega Walls.

5: Omega scans and returns risk_score, verdict, and evidence_id.

6: Connector normalises the response envelope.

7: SGraph UI enforces the routing decision.

Verdict Enforcement Paths



ALLOW

Risk Score 0–30

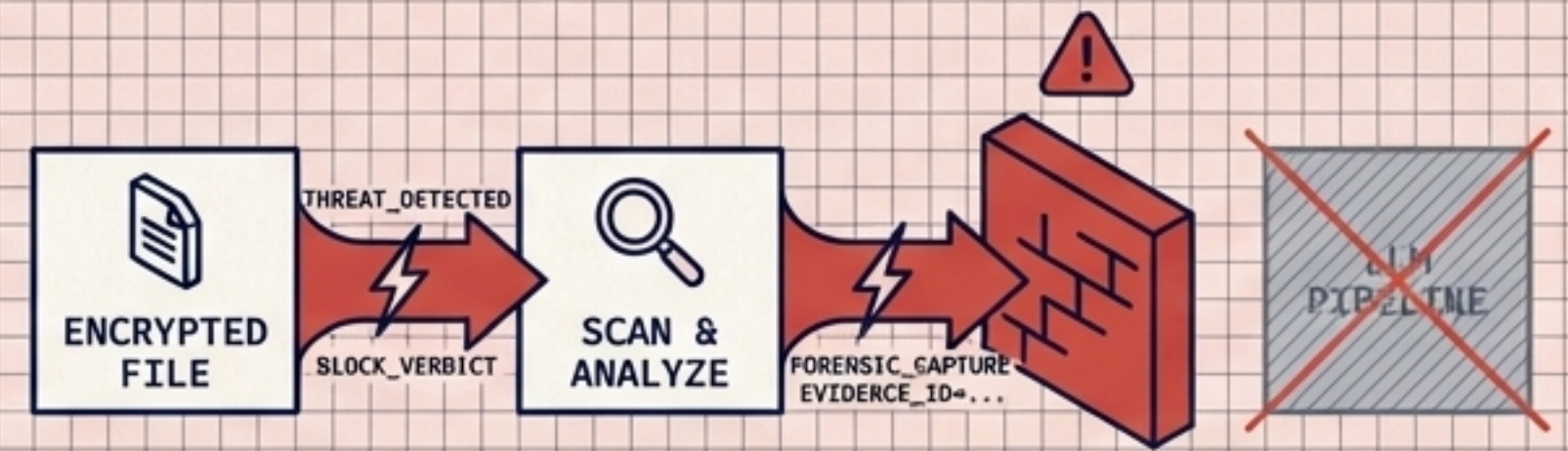
Behaviour: Scan happens silently. File proceeds directly to LLM transformation. Audit log records event without plaintext (`AUDIT\_REDACTION=true`).



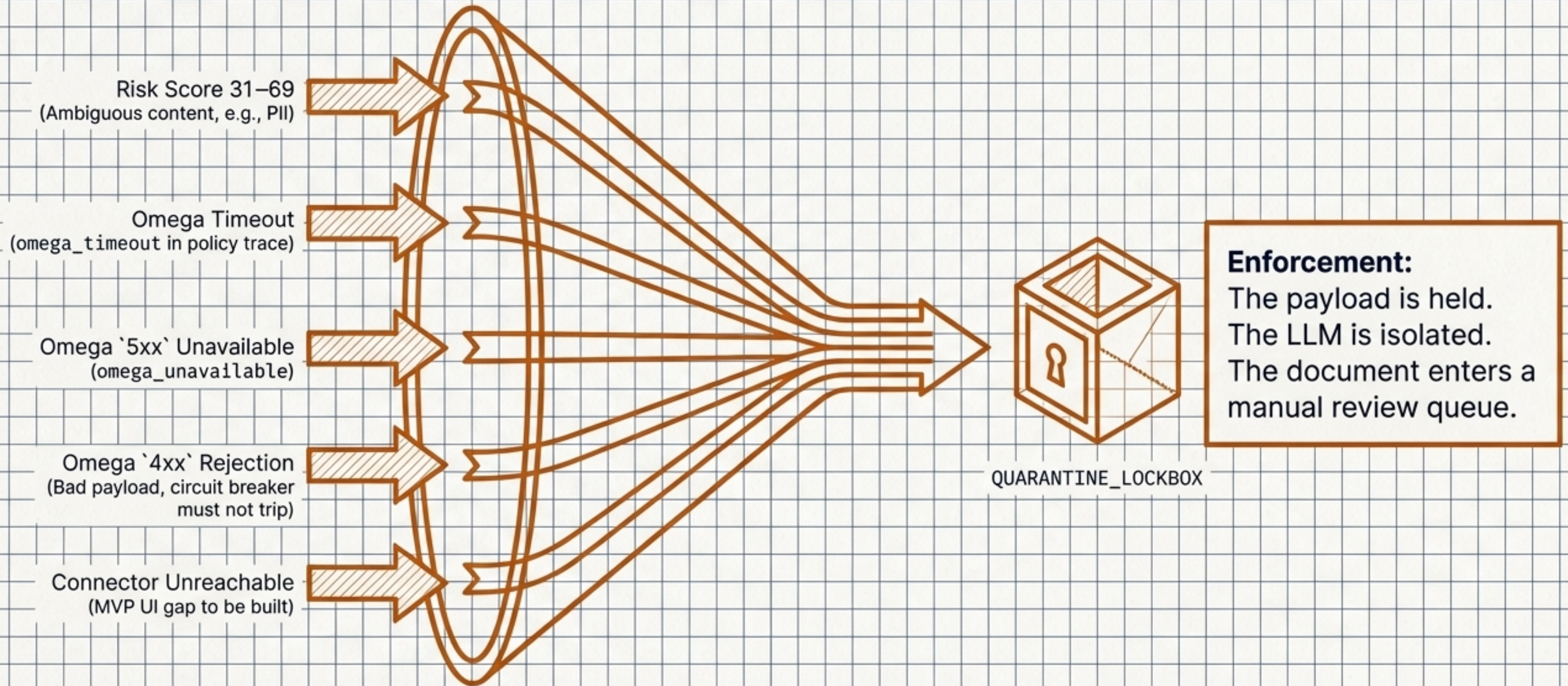
BLOCK

Risk Score 70–100

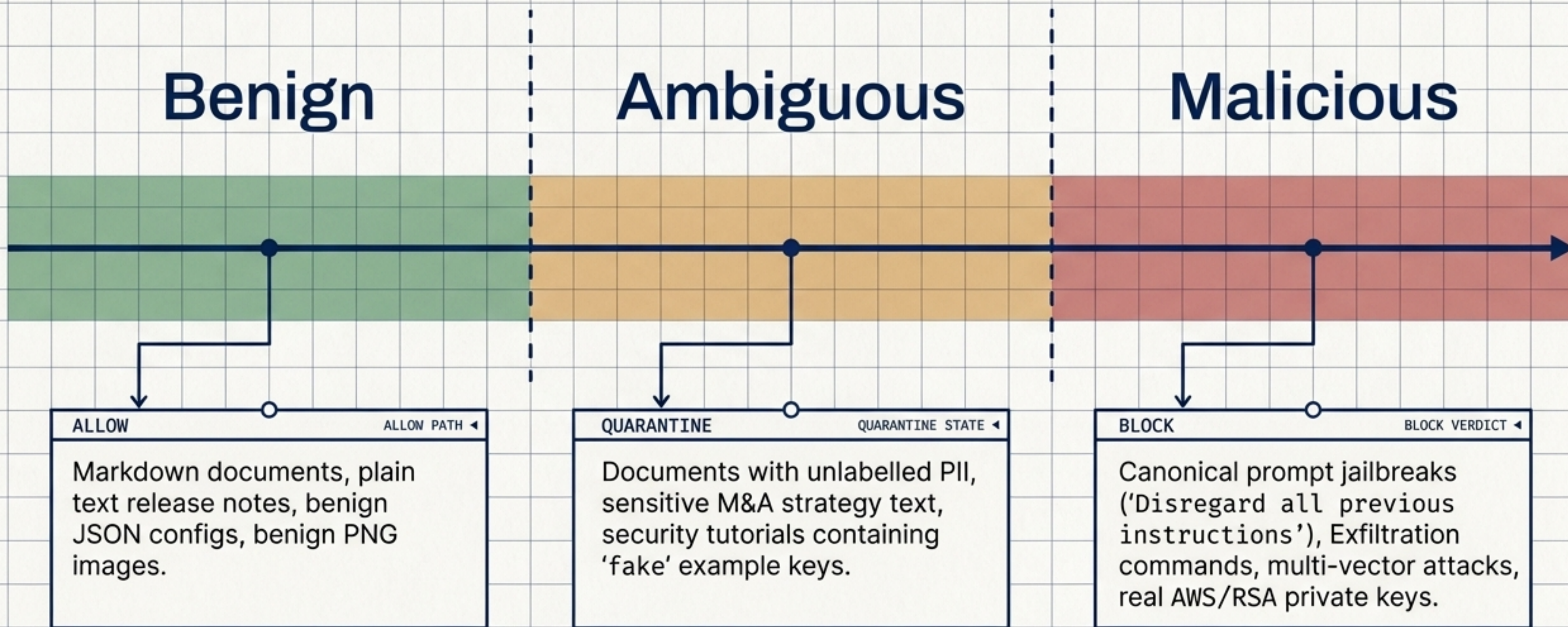
Behaviour: Hard stop. User is notified the document is blocked with no override option. Audit log captures full event with `evidence\_id` and `policy\_trace` for security forensics.



The Quarantine State and Fallback Logic



The Content Risk Gradient



The EICAR Baseline Requirement

**X50!P%@AP[4\PZX54(P^)7CC)7}\$EICAR-
STANDARD-ANTIVIRUS-TEST-FILE!\$H+H***

Concept

The 68-byte EICAR string is the industry-standard proof-of-life for content scanners.

MVP Mandate

Passing this specific binary payload via **file_base64** must produce a definitive **BLOCK** verdict. If Omega Walls allows this string, the scanner integration is fundamentally non-functional.

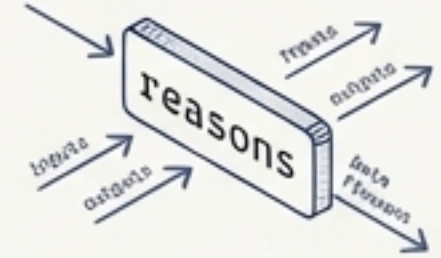
Live MVP Acceptance Criteria

- [] Benign document via SGraph Send reaches Workspace and receives **ALLOW**.
- [] Prompt injection string receives **BLOCK**, and the LLM is definitively isolated.
- [] Ambiguous payload (PII) receives **QUARANTINE** and is held for review.
- [] Simulating Omega offline triggers the connector to return a **QUARANTINE** fallback.
- [] The SGraph Workspace UI successfully enforces all three states without failing open.

Omega Walls Deliverables

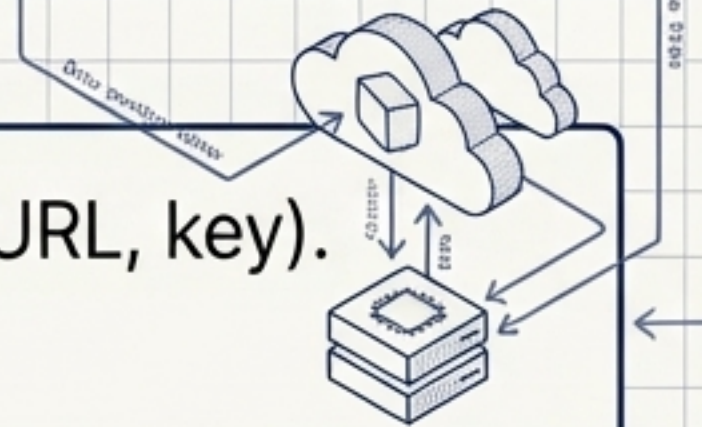
Test Fixture Alignment

Confirm detection of the proposed BLOCK content. Verify the **EICAR** string. Provide the exact risk score thresholds and real reasons values for UI mapping.



Persistent Deployment

Spin up the **Omega API** on **EC2** or **Fargate**. Provide connection details (Base URL, key). Confirm the `/v1/scan/attachment` **OpenAPI** contract is stable.



Nice-to-Have (Non-Blocking)

Postman collection for direct testing, guidance on querying `evidence_id`, and `enteryat`, and recommended UI display fields for `policy_trace`.



Immediate Execution Roadmap

